



مدارس
للتنولوجيا
التطبيقية

IT Department
Web Development and Programming

تطوير المواقع والبرمجيات
Unit 17 React.js
الصف الثالث

Web Development and Programming
Question Bank

1st.

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تطوير المواقع والبرمجيات

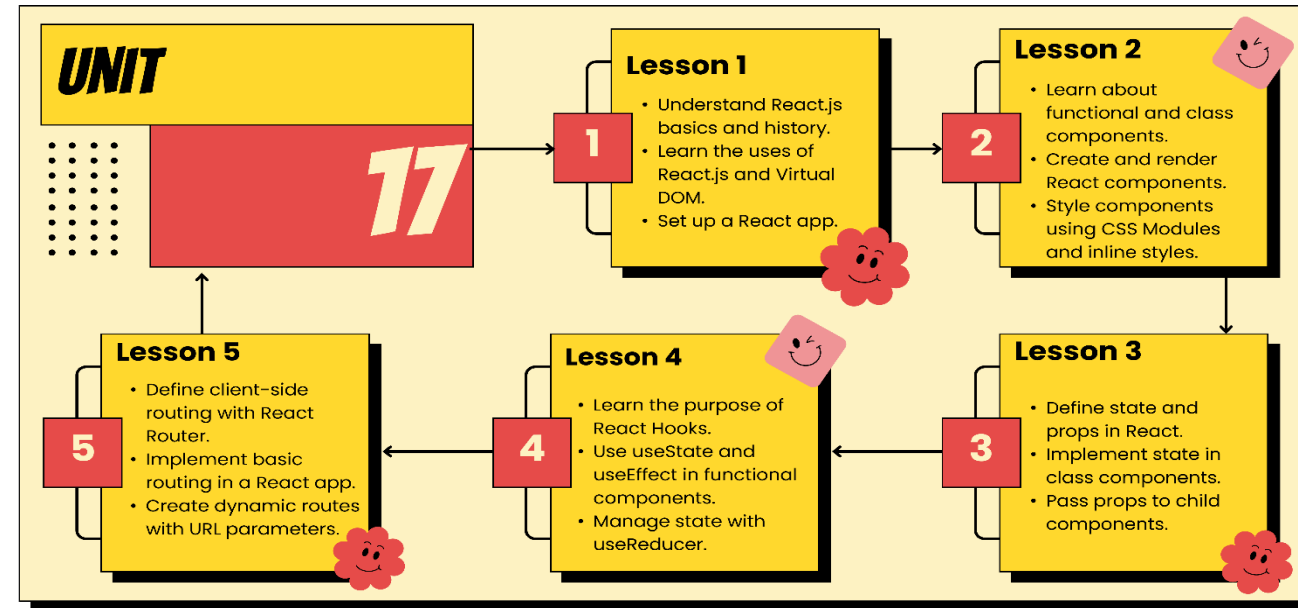
Unit 17 Reacr.js

الصف الثالث

Unit	17
Name	ReactJS
Goals / Outcomes	By the end of this Unit, students should be able to: ➤ <u>Remembering</u> 1. Recall the basic concepts, history, and uses of React.js, including the Virtual DOM, React components, state, props, hooks, and routing. 2. Identify the advantages and differences of React components, Hooks, and React Router. ➤ <u>Understanding</u> 1. Explain the principles of React, including how it integrates with the Virtual DOM and the importance of state and props in building dynamic UIs. 2. Describe the use and differences between functional and class components, state management techniques, and the purpose of different Hooks (useState, useReducer, useContext, useEffect). 3. Understand client-side vs. server-side rendering and how React Router manages routing in single-page applications. ➤ <u>Applying</u> 1. Set up a React development environment and create basic React applications using create-react-app. 2. Implement and manage state in functional and class components using useState, useReducer, and useContext. 3. Create and render React components, establishing parent-child relationships and passing data through props. 4. Use React Router to create a multi-route single-page application with navigation and dynamic routes. 5. Apply various styling methods (CSS Modules, inline styles, JavaScript objects) and integrate third-party libraries like Bootstrap. 6. Utilize useEffect for managing side effects and component lifecycle events. ➤ <u>Analyzing</u> 1. Analyze the role of the Virtual DOM in performance, and compare functional and class components. 2. Evaluate the differences and appropriate use cases for useState vs. useReducer. 3. Compare the advantages and implications of using SSR vs. CSR in React applications. 4. Examine the impact of different routing and state management techniques on application behavior and structure. ➤ <u>Evaluating</u> 1. Assess the suitability and impact of using React.js, Hooks, and React Router for different types of projects. 2. Evaluate the benefits of various state management approaches and the use of useContext for sharing state across components.

	3. Judge the effectiveness of React Router in managing single-page applications and the pros and cons of SSR and CSR. ➤ Creating 1. Build a dynamic React application that utilizes components, state, props, and Hooks for interactivity and state management. 2. Design complex component hierarchies with effective state sharing using useContext. 3. Develop a single-page application with multiple routes, dynamic navigation, and SSR using React Router. Implement a state-sharing mechanism and lifecycle management using various Hooks (useState, useReducer, useContext, useEffect).	
Knowledge	Code	Knowledge
	TPK20	writing asynchronous code using various techniques
	TPK21	Basic routing and calling a controller method from a route
	TPK22	Analyze and solve common web applications tasks by writing PHP programs
Skill	Code	Skill
	TPC5.4	Use database software tools to run queries and produce reports
	TPC5.5	Write PHP scripts to handle HTML forms
	TPC5.7	Develop and use simple Dashboards and simple data visualization forms

Unit Preface



المعرفة	الكود
writing asynchronous code using various techniques	TPK20

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
	ReactJS is a back-end JavaScript framework.	
	React was developed by Facebook.	
	React is primarily used for building single-page applications.	
	React uses a virtual DOM to improve performance.	
	React compares the virtual DOM with the real DOM to determine changes.	
	The virtual DOM updates the entire real DOM on every state change.	
	Virtual DOM improves performance by minimizing direct DOM manipulations.	
	Node.js is required to run a React application.	
	Vite is the only tool you can use to create React apps.	
	npm create vite@latest is the command to start a new Vite project.	
	You must restart the development server after every file change.	
	The App.jsx file is the root component in a new React project.	
	React components can only return one root element.	
	JSX stands for JavaScript XML.	
	JSX is executed directly by browsers.	

Dear learner: Choose the correct answer.

	What is ReactJS primarily used for? a) Backend development b) Building user interfaces c) Database management d) Mobile app development	
	ReactJS is maintained by: a) Google b) Microsoft c) Facebook d) Amazon	
	Which feature makes ReactJS efficient for UI updates? a) Virtual DOM b) Global CSS c) Server-side rendering d) Inline styles	
	What syntax extension does React use for templating? a) HTML b) CSS-in-JS c) JSX d) TypeScript	
	React components are: a) Non-reusable b) Reusable c) Only for class components d) Limited to 10 per app	
	The Virtual DOM is a: a) Real DOM copy b) Browser plugin c) Lightweight in-memory representation d) State management tool	
	How does Virtual DOM improve performance? a) By updating the entire DOM b) By reducing component count	

	c) By disabling CSS d) By minimizing direct DOM manipulations	
	What triggers a Virtual DOM update? a) State/props changes b) CSS changes c) User login d) API calls	
	To create a React app with Vite, run: a) npm create-react-app b) npm create vite@latest c) yarn add react d) npx install react	
	After creating a Vite project, you must run: a) npm run dev b) npm start c) npm build d) npm deploy	
	The default port for Vite is: a) 3000 b) 5173 c) 8080 d) 8000	
	Vite supports frameworks like: a) Only React b) Angular only c) React, Vue, Svelte d) Django	
	JSX stands for: a) JavaScript XML b) Java Syntax Extension c) JSON XML d) JS Extensions	
	JSX allows embedding expressions with: a) Parentheses ()	

	b) Angle brackets <code><></code> c) Square brackets <code>[]</code> d) Curly braces <code>{}</code>	
	To render multiple JSX elements, wrap them in: a) <code><div></code> b) <code><Fragment></code> or <code><></code> c) <code></code> d) <code><section></code>	

Dear learner: Give signific term for question.

	The virtual representation of the actual DOM that React uses for efficient updates.	
	The syntax extension for JavaScript that allows writing HTML-like code in React.	
	The React method used to render components to the DOM in older React versions.	
	The modern build tool that's faster than Create React App for setting up React projects.	
	The self-contained, reusable building blocks of a React application.	
	The type of component defined as a JavaScript function.	
	The type of component that uses ES6 classes and extends <code>React.Component</code> .	
	The React object that holds mutable data specific to a component.	
	The immutable data passed from parent to child components.	
	The React Hook that manages state in functional components.	

	The problem occurs when passing props through multiple nested components.	
	The React Hook for managing complex state logic with reducers.	
	The React Hook that shares data across components without prop drilling.	
	The React Hook for performing side effects in functional components.	
	The popular CSS framework that can be added to React via CDN or npm.	
	The utility-first CSS framework that can be configured with Vite.	

Match the correct answer from column A what suits column B

No	Column (A)	No.	Column (B)	Answer
1	Virtual DOM	A	Hook for complex state logic with reducers	
2	JSX	B	Hook for side effects (e.g., data fetching)	
3	useState	C	A. Lightweight in-memory representation of the actual DOM	
4	useEffect	D	Hook for accessing React context globally	
5	useReducer	E	HTML-like syntax extension for JavaScript	
6	useContext	F	Hook for managing state in functional components	

Dear learner: Complete the question.

	React.js is primarily used for creating user interfaces for _____ applications.	
	The main concept behind React's performance optimization is the _____ DOM.	
	React components can be either _____ or class-based.	
	JSX allows you to write _____-like code within JavaScript.	
	To manage state in functional components, you use the _____ Hook.	
	Props in React are _____ (cannot be modified by child components).	
	The tool recommended for creating modern React apps is _____.	
	For complex state logic, the _____ hook is preferred over useState.	
	Bootstrap can be added to React via CDN or _____ manager.	
	Tailwind CSS is configured in Vite through the _____ file.	
	Inline styles in React are written as _____ objects.	
	Class components manage state using this. _____.	
	To share global data without prop drilling, you use the _____ hook.	

المعرفة	الكود
Basic routing and calling a controller method from a route	TPK21

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

	The useReducer hook takes a reducer function and an initial state value as arguments.	
	When using useReducer, you call a dispatch function with an action object to update the state.	
	The useContext hook is primarily used for managing state in deeply nested components without prop drilling.	
	useEffect is a React Hook that lets you perform side effects in class components.	
	Fetching data from an API is an example of a side effect that useEffect can handle.	
	React Router is part of the core React library.	
	BrowserRouter uses the HTML5 history API.	
	Link components prevent full page reloads.	
	useParams can access route parameters.	
	Nested routes require the Outlet component.	

Dear learner: Choose the correct answer.

	React Router enables: a) Server-side routing b) State management c) Database queries d) Client-side navigation	
	To define routes, use: a) <Routes> and <Route> b) <Link> and <NavLink>	

	c) <div> and d) <BrowserRouter> only	
	<Outlet /> is used for: a) Nested routes b) State management c) API calls d) Styling	
	useNavigate() returns a: a) State variable b) Route component c) Function for programmatic navigation d) CSS class	
	A 404 page uses the route path: a) "/404" b) "*" c) "/error" d) "not-found"	
	Inline styles in JSX use: a) CSS strings b) HTML classes c) JSON files d) JavaScript objects	
	Tailwind CSS is a: a) JavaScript library b) Utility-first CSS framework c) State manager d) Backend tool	
	Global CSS files are imported in: a) App.jsx b) main.jsx or index.js c) package.json d) vite.config.js	
	To install Tailwind, run: a) npm install tailwindcss	

	b) npm install react-tailwind c) yarn add bootstrap d) npx create-tailwind	
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Match the correct answer from column A what suits column B

No	Column (A)	No.	Column (B)	Answer
1	<BrowserRouter>	A	Hook to extract dynamic URL parameters	
2	<Routes>	B	Client-side navigation component	
3	<Route>	C	Top-level component for client-side routing	
4	<Link>	D	Placeholder for nested route content	
5	useParams()	E	Container for <Route> definitions	
6	<Outlet />	F	Maps a URL path to a component	

Dear learner: Give signific term question.

	A utility-first CSS framework for rapid UI development.	
	The library that enables client-side navigation in React applications.	
	The top-level component that syncs your app with the browser's URL.	
	The component that defines the mapping between URLs and components.	
	The container component that holds multiple Route definitions.	

	The component used for navigation that prevents full page reloads.	
	The special Link component that applies styling when active.	
	The hook that extracts dynamic parameters from the URL.	
	The hook used for programmatic navigation between routes.	
	The special component that renders matched child routes in nested routing.	

Dear learner: Complete the question.

	To install React Router, you use the package _____.	
	The top-level component for routing in React is _____.	
	Dynamic route parameters are accessed using the _____ hook.	
	For programmatic navigation, you use the _____ hook.	
	A 404 page is implemented using the route path _____.	
	Nested routes render child components using the _____ component.	
	The _____ hook is used for side effects like data fetching.	

المعرفة	الكود
Analyze and solve common web applications tasks by writing PHP programs	TPK22

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

	You can embed JavaScript expressions in JSX using curly braces {}.	
	JSX fragments are written as <>...</>.	
	React components must always start with a capital letter.	
	Functional components can't use state.	
	Class components are the modern preferred way to write components.	
	Components can return other components.	
	Props are immutable within child components.	
	State can only be used in class components.	
	setState() is asynchronous in class components.	
	Props can only pass string values.	
	State changes always trigger a re-render.	
	Hooks can only be used in functional components.	
	useState returns the current state and a setter function.	
	You can call hooks inside loops or conditionals.	
	useEffect can return a cleanup function.	

Dear learner: Choose the correct answer.

	Class components require: a) extends React.Component b) useState() c) function keyword d) props keyword	
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	Props are passed to child components as: a) Mutable objects b) Immutable values c) CSS classes d) State variables	
	To share state between components: a) Use global variables b) Copy-paste code c) Lift state up d) Avoid props	
	State is managed using: a) useState or useReducer b) useEffect c) useContext d) useRef	
	Props are used to: a) Modify parent state b) Pass data to children c) Replace Redux d) Style components	
	Prop drilling refers to: a) Passing props through multiple layers b) Using useEffect c) Styling components d) Server-side rendering	
	To avoid prop drilling, use: a) useState b) Class components c) Inline styles d) Context API or Redux	
	Rules of Hooks include: a) Call Hooks in loops b) Call Hooks at the top level	

	c) Use only in class components d) Avoid useState	
	useReducer is used for: a) Simple state b) Styling c) Complex state logic d) Prop drilling	
	useContext helps with: a) Global state management b) Local state only c) API calls d) Inline CSS	
	useEffect replaces lifecycle methods like: a) componentDidMount b) render c) constructor d) shouldComponentUpdate	

Match the correct answer from column A what suits column B

No	Column (A)	No.	Column (B)	Answer
1	State	A	Handle side effects (e.g., API calls)	
2	Props	B	Sharing state across components without prop drilling	
3	useReducer	C	Mutable data local to a component	
4	useContext	D	Passing props through multiple layers (anti-pattern)	
5	useEffect	E	Immutable data passed from parent to child	
6	Prop Drilling	F	Managing complex state transitions	